



Bike Store End to End Project

A complete Business Intelligence journey — from raw data to actionable insights.

Business Intelligence Lifecycle

1

Raw Data

Excel dataset ingestion and preparation

2

Data Warehouse

Bronze, Silver, and Gold layers

3

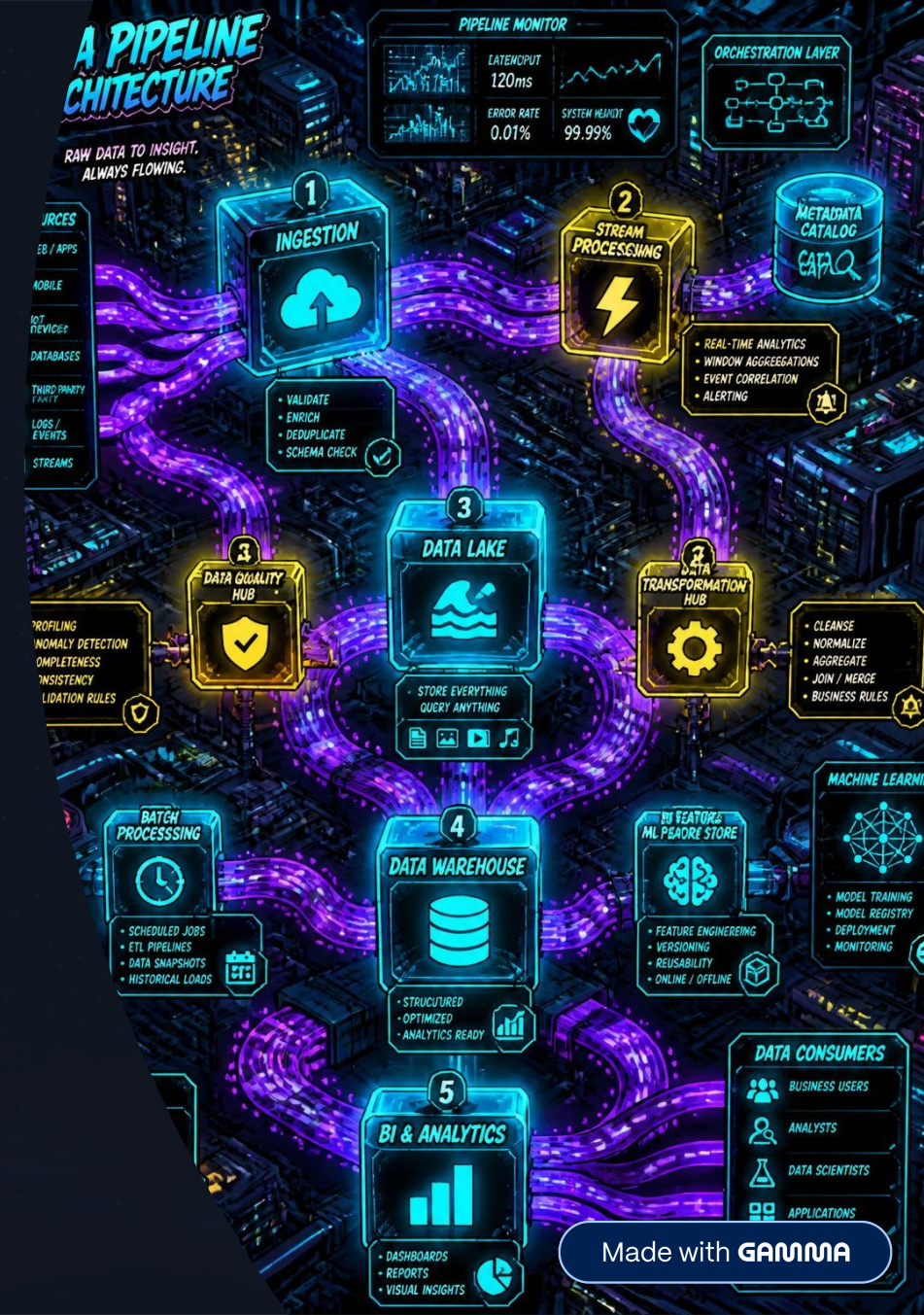
Data Modeling

Star Schema and DAX Measures

4

Dashboards & Insights

Power BI reports and business decisions



Data Modeling Fundamentals



drawSQL
Database Schema

Primary Key Integer Variable Character Date Decimal

Dimensional Model Design

Optimized for analytical queries and fast aggregations

Fact & Dimension Tables

Fact tables hold transactional data; Dimension tables provide descriptive context

Keys & Relationships

Primary Keys, Foreign Keys, and One-to-Many Relationships defined

Date Dimension

Enables time intelligence and temporal analysis in Power BI

Star Schema Diagram

The Star Schema connects a central Fact Table to surrounding Dimension Tables, enabling efficient slicing and dicing of sales data across customers, products, stores, and time.



Data Warehouse Architecture

Bronze Layer

Raw Ingestion

Data loaded as-is from source Excel files — no transformations applied. Preserves original fidelity for auditability.

Silver Layer

Cleaned & Transformed

Data cleaning, standardization, deduplication, and type casting applied. Ready for modeling.

Gold Layer

Analytics-Ready

Refined, aggregated, and optimized data serving dashboards, reports, and business insights.

Data Warehouse Layers

The layered architecture ensures data quality and traceability at every stage of the pipeline.



Executive Sales Dashboard

\$7.98M

Total Revenue

1,445K

Total Customers

4,214K

Total Orders

\$6K

Avg Order Value

The executive dashboard surfaces Sales by Category, Monthly Revenue Trends, and Top Performing products and staff — all in a single Power BI view.



Customer & Product Analytics



What the Analytics Reveal

- **Top 10 Customers by Sales** — identify high-value accounts for retention focus
- **Customer Distribution by State** — geographic demand mapping
- **Top 10 Products by Volume** — best-sellers driving revenue
- **Sales by Brand & Category** — portfolio performance at a glance
- **Store Performance & Staff Productivity** — operational efficiency metrics



Business Insights & Recommendations



High-Value Customer Retention

Identify top customers by lifetime value and design targeted loyalty programs to reduce churn and increase repeat purchases.



Optimize Product Mix

Leverage sales performance data to prioritize high-margin SKUs, phase out underperformers, and align inventory with demand signals.



Data-Driven Operations

Use store performance and staff productivity metrics to make informed decisions on inventory replenishment and workforce allocation.